

Soft Jellyfish Propulsion Simulation Midterm Report

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Abstract—This report details progress made in the use of numerical methods for modeling soft robotic jellyfish propulsion and presents preliminary results obtained using a discrete elastic rod (DER) simulation. We implement a 2D, neutrally buoyant 5-node elastic beam with time-varying natural curvature to replicate jellyfish bell contractions, and we model hydrodynamic interactions using low-Reynolds-number Resistive Force Theory. Early simulation results show that while the bell undergoes cyclic deformation, the model produces negligible vertical displacement due to the simplified hydrodynamic assumptions. These findings highlight both the feasibility of using DER for soft bell kinematics and the limitations of low-order fluid modeling when predicting forward propulsion.

I. PROBLEM STATEMENT AND MOTIVATION

Marine robots are becoming increasingly valuable to scientific study and commercial exploration, but remain limited in lifetime and range by propulsion sources that can be inefficient and hazardous to fragile sea-life [1]. As the field of soft robotics has matured, biomimetic designs including those inspired by jellyfish propulsion present a promising route to low-energy, highly maneuverable underwater platforms.

Jellyfish swim by producing coordinated bell contractions that generate vortex rings and jet thrust, achieving high efficiencies at low speeds [2]. This proposal seeks to evaluate the applicability of a discrete differential geometry model toward reproducing these fluid–structure interactions in simulation, which could decrease computational overhead for predicting the propulsive performance of soft robotic designs. Previous experimental and computational work on jellyfish-inspired robots and live-animal augmentation provides benchmarks and modeling motifs for our simulation-based exploration [4], [5].

II. INITIAL APPROACH

The jellyfish is modeled as a neutrally buoyant discrete elastic beam immersed in a viscous fluid. The simulation presented here implements a 2D beam consisting of five nodes connected by stretching springs and bending springs derived from DER energy formulations. Natural curvature is computed directly from the initial parabola-shaped bell. The initial parabolic nodal positions are defined parabolically according to Equation 1 for an n -node system.

$$\begin{cases} x(t) = wd * (\frac{t}{n-1} - \frac{1}{2}) \\ y(t) = -(ht * (\frac{2t}{n-1} - 1))^2 + y_{ctr} \end{cases} \quad t \in [0, n-1] \quad (1)$$

Time-varying curvature is then added as a periodic function of time to emulate the bell’s power stroke and recovery stroke. Curvature amplitude is highest near the center and

decreases toward the edges, with a delay term applied to capture the observed phase lag between central and peripheral contraction.

Hydrodynamic loading is modeled using low-Reynolds-number Resistive Force Theory (RFT). Drag forces proportional to normal and tangential components of local velocity are applied at each node through a Voronoi-weighted formulation [6]. These forces oppose motion but do not capture unsteady vortex generation or momentum jetting.

Time integration is performed using an implicit Newton–Raphson solver at each time step, combining inertial, elastic, viscous, and hydrodynamic contributions.

III. BACKGROUND AND EXISTING LITERATURE

A. Actuated Robotic Jellyfish: IPMC and SMA approaches

Najem et al. developed a biomimetic jellyfish-inspired underwater vehicle actuated by ionic polymer metal composites (IPMCs), demonstrating fabrication strategies and characterizing swimming performance for flexible-actuator-driven medusan robots. Their work shows that soft flexible actuators can reproduce bell-like contractions and serve as a practical benchmark for achievable kinematics in low-Reynolds number experimental platforms [2].

Earlier work at Virginia tech presented a jellyfish robot actuated by shape-memory-alloy (SMA) composite actuators. The robot, named “Robojelly” used SMA wires in parallel with spring steel spines to achieve large changes in natural curvature with magnitudes comparable to organic muscles. As shown in Figure 1, separate versions of the robot were built and characterized; one with a fully intact bell and one with “segmented” flaps around the bell margin. These impact of this change in bell shape could also be a target for modeling by our simulation. The study further emphasizes fabrication, kinematics replication, and experimental measurements of swimming performance, providing additional validation points for actuator timing and amplitude in biomimetic designs [1].

B. Mechanical Design and Motion Control

Although it revolves around a mostly rigid robot, the work presented at the Chinese Control Conference (Li & Yu) discusses mechanical structure-driven robotic jellyfish designs, motion analysis, and simulation of control schemes for steering and thrust modulation. An actuated ballast was incorporated to vary the center of the bell for bioinspired orientation control [3]. This study indicates the value of coupling mechanical design choices to control strategies to achieve desired swimming behaviors and maneuverability in robotic jellyfish.



Fig. 1: Images of Robojelly with unsegmented (top) and segmented (bottom) bell margin [1].

C. Computational FSI and Morphological Effects

Miles & Battista used IB2d, an open-source implementation of the immersed boundary mathematical framework, to study how tentacles and oral arms affect locomotion and found that appendages can strongly influence vortex formation and forward speed. In many parameter regimes, increased density or length of tentacles or oral arms were found to significantly decrease forward speed and change energy dissipation/mixing behavior [4].

D. Biological Augmentation and Application Context

Recent work by Anuszczyk & Dabiri demonstrates electromechanical stimulation of live jellyfish muscle tissue for ocean exploration applications, showing the potential for hybrid bio-robotic systems and reinforcing the broader interest in jellyfish-inspired platforms for low-power ocean monitoring. Their discussion provides context for realistic performance goals and operational use-cases for jellyfish-inspired vehicles [5].

IV. SIMULATION MODEL IMPLEMENTATION

The implemented numerical simulator follows a discrete elastic rod formulation. Stretching forces and bending forces are computed from analytical gradients and Hessians of DER energy terms. Natural curvature is determined from the initial bell geometry using a discrete curvature operator based on adjacent tangent vectors.

A time-varying curvature function modulates this natural curvature, producing contraction and relaxation phases across each cycle. The activation pattern incorporates amplitude gradients, temporal delays, and exponential decay during recovery. It is defined as a piecewise function of cycle time by Equation 2.

$$\Delta\kappa(t) = \begin{cases} \Delta\kappa_{max} * \frac{t - t_{del}}{0.3P - t_{del}} & t \in [0, 0.3P) \\ \Delta\kappa_{max} * e^{-\frac{t - 0.3P}{\tau_{relax}}} & t \in [0.3P, P) \end{cases} \quad (2)$$

Hydrodynamic forces are calculated via Resistive Force Theory, using tangential and normal drag coefficients and Voronoi-segmented nodal lengths. The model assumes low Reynolds number flow and does not include vortex dynamics or added-mass effects.

Newton–Raphson iteration is used to solve the implicit Euler method of integration for the nonlinear equations of motion at each time step, combining inertial, elastic, hydrodynamic, and external contributions.

V. SIMULATION PARAMETERS AND SETUP

The bell is initialized as a parabola spanning 0.1 m in width and 0.08 m in height. Five nodes discretize the geometry. Material and physical parameters include bending stiffness $EI = 0.01$, stretching stiffness $EA = 30$, fluid drag coefficients $C_t = 0.15$ and $C_p = 0.30$, and a beam radius of 0.01 m with density 1050 kg/m^3 . The simulation runs for 5 s with a time step of $\Delta t = 0.005 \text{ s}$.

Natural curvatures are computed from the resting shape, and time-dependent curvature is applied using a 2 s period, with a strong central contraction ($\Delta\kappa = -1.5$) and weaker edge actuation. A delay parameter governs the phase lag across interior nodes.

Outputs tracked include node positions, velocities, and lateral and vertical displacements of the end nodes and center node.

VI. RESULTS AND OBSERVATIONS

The time variant natural curvature approach to modeling jellyfish contractions was successful in producing reciprocal motion of the bell. Plots of the beam’s shape over the duration of a single cycle are shown in Figure 2. For each contraction cycle, the simulated jellyfish bell is initially parabolic in shape, but undergoes a shape change where natural curvature quickly increases to a peak value, resulting in the edges of the bell drawing closer together before slowly relaxing back to resting. While it is unclear if this approach to kinematic modeling is accurate to a real jellyfish, it should provide a strong analogue for actuators used in jellyfish robots, like IPMC and BISMAL actuators, which also alter the bell’s natural curvature to produce motion [1], [2].

Although the vertical positions of nodes shifted during the course of each individual cycle, the simulation failed to produce net movement of the structure over the full duration. The plot in Figure 3 shows how upward movement of the apex node during bell contraction was canceled by equal downward movement during the relaxation phase, resulting in no net progress. This finding aligns with the scallop theorem, which explains that propulsion in low-Reynolds environments is not possible through geometrically reciprocal motion, even if the timescales are different for contraction and relaxation [7].

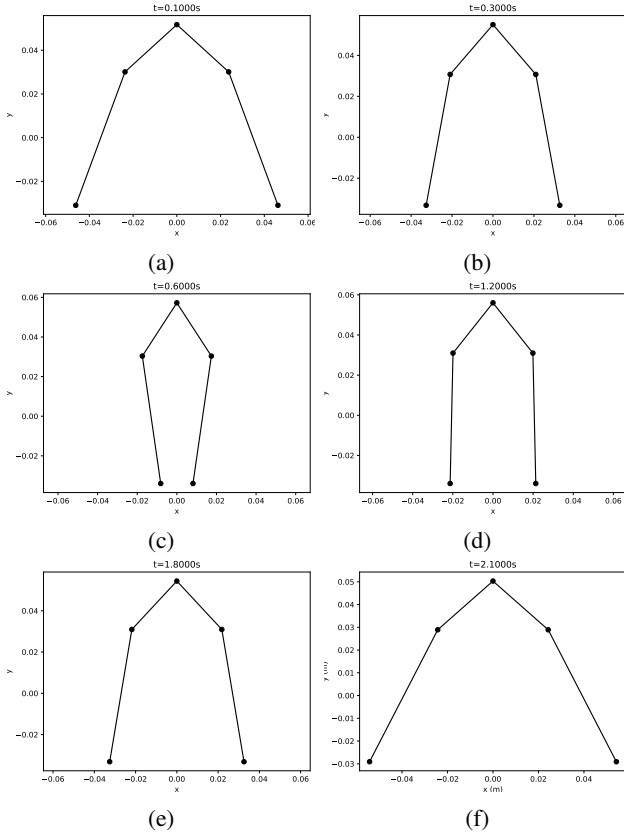


Fig. 2: System states at 0.1 (a), 0.3 (b), 0.6 (c) (maximum contraction), 1.2 (d), 1.8 (e), and 2.1 (f) (start of next cycle) seconds of simulation.

In addition to the vertical position of the apex node, horizontal positions of the bell’s edges were also tracked during the simulation. Figure 4 plots the x-coordinates of the first and last nodes, which converge toward zero in each cycle as the curvature rises before gradually relaxing back to their starting points. Also visible in these plots is how changes in the direction of travel appear to generate oscillations in position which are not present in real world jellyfish or jellyfish robots, suggesting the damping ratio of our simulation was less than that of real world bells due to some combination of inaccuracies in material properties, hydrodynamic forces, and nodal mass.

VII. LIMITATIONS AND FUTURE WORK

The initial formulation had several shortcomings that must be addressed before it can inform research findings. Chief among them is the lack of progress of the structure in any direction. To escape the effects of the scallop theorem we must implement a more complex hydrodynamic model that includes phenomena of higher Reynolds number environments [7]. This change is further supported by studies of real jellyfish indicating that their movement takes place in higher Reynolds flow and is dominated by jetting and vortex effects [2], [5]. Balancing simulation accuracy with computational efficiency will be crucial, as the key benefit of using a DER

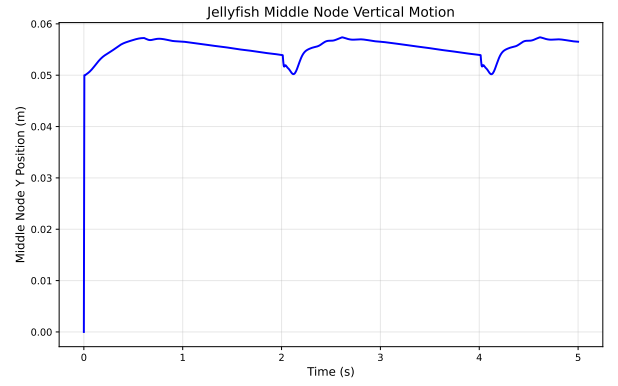


Fig. 3: Y-position of the jellyfish apex node as a function of time.

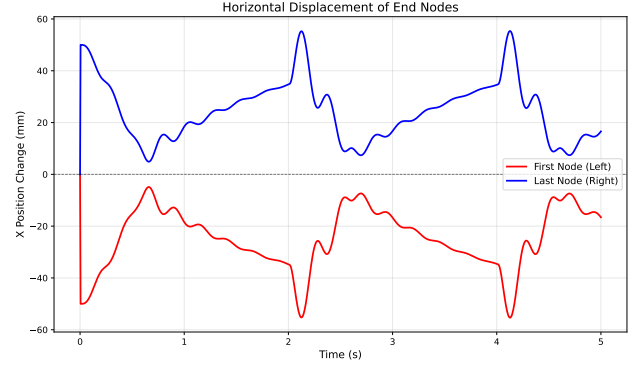


Fig. 4: X-positions of the jellyfish edge nodes as functions of time.

formulation is its relative speed compared to finite element methods [6].

Another goal for future work is finding more realistic values for structure stiffness, mass, and drag coefficients. This should not only help eliminate inaccurate oscillatory motion, but also improve the trustworthiness of the model and its utility as a comparative tool.

Finally, once the model is more representative of a real jellyfish and its surrounding fluid structure interactions we can experiment with the control of its actuation. While the current activation function is based on intuition and observation of biological jellyfish movements, future functions could attempt to optimize thrust relative to energy input or investigate differences between jetting and rowing swimming modes [2].

VIII. CONCLUSION

We implemented a discrete elastic rod simulation with time-varying natural curvature to model jellyfish bell contractions. The framework successfully reproduces cyclic bell deformation, but due to the low-Re hydrodynamic model generates no vertical displacement or thrust. These findings validate the mechanical portion of the model while motivating future work to incorporate more realistic fluid–structure interaction.

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